

Recognizing an Adult Speaker's Sex from Voice

Early Development of Sex Recognition Abilities

- The ability to discriminate male from female voices emerges early in human development.
- Infants as young as six months old can consistently distinguish male from female voices.
- Children between six and nine years can identify the gender of an adult's voice from a spoken vowel (>98% correct) or whispered vowel (about 79% correct) with accuracy comparable to adults.
- This early development suggests the fundamental importance of sex recognition from voice in human communication and social development.

Robustness of Sex Recognition Across Stimulus Types

- Sex recognition from voice remains robust even under conditions of acoustic distortion, consistent with the multiplicity of cues available to listeners.

Stimuli	Recognition accuracy	References
Voiceless front fricative (f, th)	55%–71.5%	Ingemann (1968); Schwartz (1968)
Voiceless central fricative (s, sh)	72.5%–91.5%	Ingemann (1968); Schwartz (1968)
Whispered vowel	75%–97.5%	Ingemann (1968); Lass et al. (1976); Schwartz and Rine (1968)
2 cycles of phonation	appx. 80%–90%	Owren et al. (2007)
Low-pass filtered vowel	91%	Lass et al. (1976)
Vowel	96%–100%	Lass et al. (1976, 1979)
Monosyllabic word	99%	Lass et al. (1979)
Bisyllabic word	95%	Lass et al. (1979)
Low-pass filtered sentence	97%	Lass et al. (1980)
Backward sentence	99%	Lass et al. (1978)
Sine wave replica of a sentence	≈ 83%	Fellowes, Remez, and Rubin (1997)
Sentence	98%–100%	Lass et al. (1979, 1980)

Effect of Stimulus Length and Complexity on Recognition Accuracy

- As stimuli increase in length and complexity, they contain more (and more kinds of) acoustic information, including pitch contours and more details about pronunciation.
- Accuracy of sex recognition increases accordingly, with stimuli as long as a syllable consistently yielding near 100% accuracy.
- Even heavily degraded stimuli (like sine wave replicas) or reversed speech maintain high recognition rates, demonstrating the robustness of sex cues in the speech signal.

The Male Recognition Advantage and Response Bias

- Several studies have reported that male voices are "easier to recognize" than female voices, with listeners typically assigning male voices to the correct category more often than female voices
- However, this may represent a response bias rather than a true perceptual difference.
- An exaggerated example of a listener who responds "male" to every sample, achieving 100% accuracy for male voices but 0% for female voices.

Signal Detection Theory and Sex Recognition

- Signal detection theory provides statistical methods for estimating listeners' perceptual acuity independent of response biases.
- This framework categorizes responses as hits, misses, false alarms, and correct rejections. Examining the ratio of hits to false alarms provides a better measure of performance than looking at accuracy for either sex in isolation. This approach allows researchers to determine whether the male recognition advantage is due to bias or reflects a true perceptual difference in the detectability of male voice characteristics.

Primary Cues to Speaker Sex

- When asked to describe stereotypically male and female voices, listeners expect male voices to be deep, demanding, and loud, and female speakers to have good enunciation, high pitch with substantial dynamic variation, and a fast and variable speaking rate (Kramer, 1977).

<i>Cue</i>	<i>Female Speech</i>	<i>Male Speech</i>
Mean F0	Averages about 220 Hz	Averages about 110 Hz
F0 dynamics	More variable; more rising pitch contours	More monotone; more falling pitch contours
Vocal tract resonances	Higher in frequency	Lower in frequency
Speaking rate	May be slower	May be faster
Articulation	More precise	Less precise

Common Misconceptions and Actual Speech Patterns

- Studies of the cues listeners use when judging a speaker's sex have focused mostly on the relative contributions of F0 and formant frequencies. Early studies separated these parameters using an electrolarynx, a device that produces a buzzing sound at a constant frequency.
- Coleman (1971) asked male and female speakers to read a passage using an electrolarynx with a steady F0 of 85 Hz. Male speakers were easily identified, but female voices were correctly identified only 79% of the time, suggesting that the low F0 biased listeners toward "male" responses despite female formant patterns. A subsequent study (Coleman, 1976) revealed that F0 alone is also inadequate to unambiguously specify a speaker's sex. When combining either artificial source (120 Hz or 240 Hz) with the opposite vocal tract, the resulting voice typically sounded male, suggesting that while F0 may be the main cue, it is neither the only cue nor sufficient when contradictory information is present.

Common Misconceptions and Actual Speech Patterns

- More recent studies have applied speech synthesis rather than electrolarynges to mix and match male and female vocal characteristics.
- Whiteside (1998) synthesized short vowels by combining voice sources and vocal tracts from male and female speakers.
- The mean F0 was generally the best predictor of perceived sex for these hybrid stimuli, usually overriding vocal tract characteristics. However, the studies collectively demonstrate that sex recognition relies on a complex integration of multiple acoustic cues, with F0 serving as the primary but not sole determinant of perceived speaker sex.